Detailed Derivation of Posterior Mean and Variance (Expanded Completing-the-Square Steps)

Bayesian Estimation

1. Problem Setup

Let:

- $x_1, x_2, \ldots, x_n$ be independent observations
- μ be an unknown mean
- σ^2 be known variance

Assume the likelihood:

$$x_i \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$$

Assume a Gaussian prior:

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

Goal:

$$\text{Derive } p(\mu \mid x_1, \ldots, x_n)$$

2. Bayes' Theorem

Bayes' theorem for parameter estimation is:

$$p(\mu \mid x) = \frac{p(x \mid \mu)\, p(\mu)}{p(x)}$$

Since $p(x)$ does not depend on μ:

$$p(\mu \mid x) \propto p(x \mid \mu)\, p(\mu)$$

3. Likelihood Function

Because observations are independent:

$$p(x \mid \mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Ignoring constants:

$$p(x \mid \mu) \propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

4. Prior Distribution

The Gaussian prior is:

$$p(\mu) = \frac{1}{\sqrt{2\pi\sigma_0^2}}\exp\left(-\frac{(\mu-\mu_0)^2}{2\sigma_0^2}\right)$$

Ignoring constants:

$$p(\mu) \propto \exp\left(-\frac{(\mu-\mu_0)^2}{2\sigma_0^2}\right)$$

5. Forming the Posterior

Multiplying likelihood and prior:

$$p(\mu \mid x) \propto \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 - \frac{(\mu-\mu_0)^2}{2\sigma_0^2}\right]$$

Taking logarithm:

$$\log p(\mu \mid x) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 - \frac{(\mu-\mu_0)^2}{2\sigma_0^2} + C$$

6. Expanding the Quadratic Terms

First expansion:

$$\sum_{i=1}^{n}(x_i-\mu)^2 = \sum_{i=1}^{n}x_i^2 - 2\mu\sum_{i=1}^{n}x_i + n\mu^2$$

Second expansion:

$$(\mu-\mu_0)^2 = \mu^2 - 2\mu\mu_0 + \mu_0^2$$

7. Substitution and Collection of Terms

Substitute the expansions into the log-posterior:

$$\log p(\mu \mid x) = -\frac{1}{2}\left[\mu^2\left(\frac{n}{\sigma^2}+\frac{1}{\sigma_0^2}\right) - 2\mu\left(\frac{\sum_{i=1}^{n}x_i}{\sigma^2}+\frac{\mu_0}{\sigma_0^2}\right)\right] + C$$

This is a quadratic function of μ.

8. Completing the Square (Expanded)

Define:

$$A = \frac{n}{\sigma^2}+\frac{1}{\sigma_0^2}, \qquad B = \frac{\sum_{i=1}^{n}x_i}{\sigma^2}+\frac{\mu_0}{\sigma_0^2}$$

Then:

$$\log p(\mu \mid x) = -\frac{1}{2}\left[A\mu^2 - 2B\mu\right] + C$$

8.1 Factor Out A

$$A\mu^2 - 2B\mu = A\left(\mu^2 - 2\frac{B}{A}\mu\right)$$

8.2 Use the Identity

Recall the identity:

$$\mu^2 - 2a\mu = (\mu - a)^2 - a^2$$

Let $a = \frac{B}{A}$:

$$\mu^2 - 2\frac{B}{A}\mu = \left(\mu - \frac{B}{A}\right)^2 - \left(\frac{B}{A}\right)^2$$

8.3 Substitute Back

$$A\mu^2 - 2B\mu = A\left(\mu - \frac{B}{A}\right)^2 - \frac{B^2}{A}$$

9. Final Gaussian Form

Substitute into the log-posterior:

$$\log p(\mu \mid x) = -\frac{1}{2}\left[A\left(\mu - \frac{B}{A}\right)^2 - \frac{B^2}{A}\right] + C$$

Rearranging:

$$= -\frac{A}{2}\left(\mu - \frac{B}{A}\right)^2 + \underbrace{\left(\frac{B^2}{2A} + C\right)}_{\text{constant}}$$

Comparing with Gaussian form:

$$-\frac{1}{2\sigma_N^2}(\mu - \mu_N)^2$$

10. Identification of Posterior Parameters

Posterior Variance

$$\boxed{\frac{1}{\sigma_N^2} = A = \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}$$

$$\boxed{\sigma_N^2 = \left(\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}\right)^{-1}}$$

Posterior Mean

$$\boxed{\mu_N = \frac{B}{A} = \sigma_N^2 \left(\frac{\sum_{i=1}^n x_i}{\sigma^2} + \frac{\mu_0}{\sigma_0^2} \right)}$$

11. Final Posterior Distribution

$$\boxed{p(\mu \mid x_1, \ldots, x_n) = \mathcal{N}(\mu_N, \sigma_N^2)}$$

12. Key Insight

The transition from Step 7 to Step 8 is achieved by completing the square, which rewrites the quadratic expression into Gaussian form, allowing direct identification of posterior mean and variance.